MILITARY SPECIFICATION

TEST HARNESS, RADIO SET AN/ARM-46(*)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy,

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be for shop use and shall be suitable for installation on a standard service bench for control and testing of Radio Set AN/ARC-75, AN/ART- 36 and A N/A RR-66 when used in conjunction with standard test equipment. The equipment shall have provisions for captivating all equipment, mounting, and controlling units of Radio Set AN/ARC-75, AN/ART-36 and AN,ARR-66.

1.2 Classification - The Test Harness, Radio Set AN/ARM-46 (*) shall consist of the following items:

Items	Type Designation	Appl. Para.
Box, Distribution	J-1169/ARM-46	3.4.1
Mounting	MT-2323/ARM-46	3.4.2
Cable Assembly, Power Electrical	W-1 of AN/ARM-46	3.4.3
Cable Assembly, Special Purpose, Electrical	W-2 of AN/ARM-46	3.4.4
Cable Assembly, Special Purpose, Electrical	W-3 of AN/ARM-46	3.4*5
Maintenance Fixture, Amplifier	MT-2062/ARM-38	3.4.6
Maintenance Fixture, Oscillator	MT-2058/ARM-38	3.4.7
Maintenance Fixture, Amplifier	MT-2371/ARM-46	3.4.8
Maintenance Fixture, Amplifier	MT-2060/ARM-38	3.4.9
Maintenance Fixture, Amplifier-Generator	MT-2059/ARM-38	3.4.10
Cable Assembly, Special Purpose, Electrical	W-4 of AN/ARM-46	3.4, 11
Cable Assembly, Special Purpose, Electrical	W-5 of AN/ARM-46	3.4.12
Cable Assembly, Special Purpose, Electrical	W-6 of AN/ARM-46	3.4.13

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Items	Type Designation	Appl. Para.
Cable Assembly, Special Purpose, Electrical	W-7 of AN/ARM-46	3.4.14
Cable Assembly, Special Purpose, Electrical	W-8 of AN/ARM-46 (3 Each)	3.4.15
Cable Assembly, Special Purpose, Electrical	W-9 of AN/ARM-46	3.4.16
Cable Assembly, Radio Frequency	W-10 of AN/ARM-46	3.4.17
Cable Assembly, Radio Frequency	W-n of AN/ARM-46	3.4.18
Cable Assembly, Radio Frequency	W-12 of AN/ARM-46	3.4.19
Cable Assembly, Radio Frequency	W-13 of AN/ARM-46	3.4.20
Cable Assembly, Radio Frequency	W-14 of AN/ARM-46 (4 Each)	3.4.21
Cable Assembly, Special Purpose, Electrical	W-15 of AN/ARM-46	3.4.22
Cable Assembly, Radio Frequency	W-16 of AN/ARM-46	3.4.23
Maintenance Fixture, Receiver RF Amplifier	MT-2321/ARM-46	3.4.24
Maintenance Fixture, Transmitter, Preamplifier	MT-2322/ARM-46	3,4.25
Tool Kit	TK- (*)/U	3.4, 26
Adapter Kit	MK- (*)/U	3.4.27
Tuning Cover Kit	MK-(*)/ARM-46	3.4.28

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military'

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-H-15362	Harnesses; Test Bench, Avionic Equipment, General Specification for

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MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and inspection, For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment For Use With Electronic and Fire Control Systems, General Specification for
MIL-T-22163	Test Harness, Radio Set AN/ARM-38

STANDARDS

Military

MIL-STD-242	Electronic Equipment Parts

2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania,

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-T-21200. Parts selected from specifications called out in Specification MIL-T-21200 shall be limited to those values, styles, tolerances, etc. , set forth in MIL-STD-242,

(2) Nonrepairable subassemblies, as outlined in specification MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. A mean-time-to failure of more than 5000 hours should be the goal of each.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity,

3. 1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-T-21 200.

3. 1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction t e sting.

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3.2 Design and Construction

3.2,.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-T-21200 for design, construction and workmanship, unless otherwise specified herein.

3.2.2 Total Weight - The total weight of the equipment, including cables, shall be a minimum consistent with good design and shall not exceed 58.5 pounds.

3.2.3 Operating Life

3.2.3.1 Operational Stability - The equipment shall operate with optimum performance for 1000 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during normal use.

3.2.3.2 Total Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.3.3 Mean Operating Time Between Failures - The equipment shall have a minimum of 1000 hours of mean operating time between failures. A failure shall be defined as any malfunctioning which causes the equipment to fail to meet the minimum performance requirements specified for the equipment.

3.2.4 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Spacification MIL-T-21200,

3.2.5 Interchangeability - The equipment shall meet the interchangeability y requirements as defined in Specification MIL -T-21200,

3.2.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to ratio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30° ±10°C)
Altitude	Normal ground
Vibration	Non e
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 V AC and 27.5 ± 0.5 V DC

3.2.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-T-21200 for Class 3 equipment.

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3.2.11 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under standard conditions 5 minutes at extreme service conditions.

3.2.12 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-T-21200 for Class 3 equipment. The power required shall be of the following types:

A C Power Single Phase, 115V, 50 to 70 cps and 380 to 420 cps, VA

AC Power, 3 Phase, 115V, 400 cps

DC Power, 27 ± 2. V

3.2.12.1 Voltage Protection - The equipment shall not be damaged by voltages either above or below the voltages specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.2.13 Standby Provisions - The equipment shall provide for a standby position whereby rated voltage shall be applied to all tube filaments. Except for the associated filament circuits, voltages shall not be applied to other parts of the equipment.

3.3 Performance -

3.3.1 General - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified,

3.3.2 External Square Wave Signal - With a square wave signal from an external source applied to input terminals of the J-1169/ARM-46, the Distribution Box shall be capable of providing separate pulse type keying signals which will operate the AN/ARC-75 or AN/ART-36 radio set to "turnaround" and/or "carry-on" condition.

3.3.3 Turnaround Signal - The turnaround signal shall be a square wave having a minimum peak-to-peak amplitude of 30 volts centered above zero and a period of 84 milliseconds when operated into a 100, 000 ohm load.

3.3.4 Carry-On Signal - The carry-on signal shall be a square wave having a minimum peak-to-peak amplitude of 30 volts centered above zero and a period of 84 milliseconds when operated into a 100, 000 ohm load. The width of the positive portion of the pulse shall be adjustable from 14 milliseconds minimum to not more than 42 milliseconds. The carry- on and turnaround pulses shall be in phase and the trailing edge shall be coincident.

3.4 Detail Requirements -

3.4.1 Box, Distribution J-1169/ARM-46 - The distribution box shall meet the following requirements:

3.4.1.1 Function - The distribution box shall provide the necessary circuitry, jacks, switches, indicator lamps, receptacles, test points, and binding posts to facilitate the test of Radio Set AN/ARC- 75, AN/A RR-66 and AN/ART-36, and shall provide a means of mounting, holding and connecting to Radio Set Control C- 273C/AR for frequency control of the radio set under test.

3.4.1.2 Form Factor - Overall dimensions of the distribution box shall not exceed 17-5/8 inches long by 9-1/2 inches wide by 7-1/2 inches deep, The unit shall have provisions for mounting to the bottom of the top shell of a test bench existing hole pattern.

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3.4.1.3 Weight -The weight of the distribution box shall not exceed 10.0 pounds.

3.4.1.4 Contents of Front Panel - The following shall be located on the front panel of the Distribution Box J- 1169/ARM-46 and shall provide the functions specified:

(1) Test Jacks -

(a) Turnaround - Terminal for monitoring turnaround signal and for application of external turnaround signal.

(b) Receiver Test Bias - Terminal for application of negative bias voltage to receiver,

(c) A-C Data Out - Provides 2000-ohm load and faltered data output from receiver.

(d) Three Grounds - Ground terminals for accessory equipment.

(e) Carrier-On - Terminal for monitoring carry-on signal and for application of external carry-on signal.

(f) Square Wave In - Terminal for connections to external square wave signal source.

(2) Test Points -

(a) D-C Data Out - Unfiltered data output measuring point.

(b) Receiver B-plus-Receiver B plus monitoring point.

(c) Carrier Level - Permits measurement of receiver signal level.

(d) Continuous B plus - Continuous B plus voltage monitoring point.

(e) Bias - Permits measurement of negative bias voltage.

(f) Antenna Relay - Permits measurements of Radio set relay operate time.

(g) Transmit B plus - Transmit B plus voltage monitor point.

(3) Indicator Lamps -

(a) Test Power

(b) Radio Power

(c) Phase 1, phase 2 and phase 3 115 V AC, 400 cps

(d) 27.5 V DC

(4) Fuses and Fuse Holders -

(a) Phase 1, phase 2 and phase 3 115 V AC, 400 cps

(b) Phase 1, phase 2 and phase 3 115 V AC, 400 cps (Spare)

(c) 27.5 V DC

(d) 27.5 V DC (Spare)

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(5) Receptacles -

(a) 115 VAC 3 phase, 400 cps and 27.5 VDC Input Power.

(b) Receptacle for connection to AN/ARC-75, AN/A RR-66 and AN/ART-36.

(c) Antenna Cable for measurement of relay operate time.

(6) Connector -

(a) Connector to provide electrical connections to C-2736/AR

(7) Meter -

(a) Directional Coupler Output to measure transmitter forward and reflected power and VSWR.

(8) Switches -

(a) Test Power - Controls power to distribution box,

(b) Radio Power - Controls power to radio set.

(c) Carry On - Controls application of carry-on signal to radio set.

(d) R-F Power (4 position) - Switches directional coupler metering circuitry.

(e) Turnaround - Controls application of turnaround signal to radio set.

(9) Adjustments -

(a) Carrier Delay - Adjusts delay of carrier-on pulse with respect to turnaround.

(b) Calibrate - Calibrates meter for measuring transmitter power output.

(c) Scale - Calibrates meter for measuring VSWR.

3.4.1.5 Electrical Connections - Connections to external circuits located on the sides of the Distribution Box shall be provided as follows:

Reference Designation	Receptacle Type	Plug Type	Function
J 2003	AN-3102A -2 0-16P		Power Input
J 2701	Bendix PT02A-22-55P or approved equivalent		Provides connections to AN/ARC-75 or AN/ARR-66 or AN/ART-36
P 1801		Cannon K03-21-305N or approved equivalent	Provides connection to C-2736/AR

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3.4.2 Mounting MT-2323/ARM-46 - The mounting shall meet the following requirements.

3.4 2.1 Functions - The mounting shall provide facilities for mounting and holding the AN//ARC-75, AN/ART-36 and AN/A RR-66 equipment securely to the surface of a test bench. The AN/A RR-66 or AN/ART-36 equipment shall fit into the mounting without a dust cover and be secured by use of the screws normally used to hold the dust cover. Provision shall be provided for rotating the equipment about a fixed center with a facility for locking at any point.

3.4.2.2 Form Factor - Overall dimensions of the MT-2323/ARM-46 shall not exceed 14-3/4 inches long by 7- 1/2 inches wide by 22 inches deep. The unit shall have provisions for mounting to the normal working surface of a test bench using existing hole patterns.

3.4.2.3 Weight - The weight of the mounting unit shall not exceed 23.2 pounds.

3.4.2.4 Contents of Unit - The mounting shall contain a 115 V AC, 2 phase, 400 cycle blower to provide adequate air for the RT-499/ARC-75 equipment.

3.4.2.5 Electrical Connections - Connections to external circuits shall be provided in accordance with MS 21993 and/or MS 2549.

3.4.3 Cable Assembly Power, Electrical W-1 of AN/ARM-46 - This cable shall provide primary power connection to Distribution Box J-1168/ARM-46 . One end of the cable shall terminate in a type AN3106-A-20-16S connector. The other end shall have no connector since the termination will depend on primary power source. The length of the cable shall be 72 inches. The maximum allowable weight shall be O. 75 pounds.

3.4.4 Cable Assembly, Special Purpose, Electrical W-2 of AN/ARM-46 - This cable shall provide the necessary electrical connections to extend Radio Set Control C-2736/AR from Distribution Box J-1169/ARM-46 to the test bench for servicing. One end of the cable shall be terminated in a Cannon K03-21-30SN connector or approved equivalent and the other end shall terminate in a Cannon K01-21-30PN or approved equivalent. The length of the cable shall be 72 inches. The maximum allowable weight shall be 0.75 pounds.

3.4.5 Cable Assembly, Special Purpose, Electrical W-3 of AN/ARM-46 - This cable shall Provide all necessary power input and control connections between Distribution Box J- 1169/ARM-46 and AN/AM-66, or AN/ART-36. One end of the cable shall terminate in a Bendix PT06P-22-55S or approved equivalent connector and the other end shall terminate in a connector capable of mating with the front panel receptacle of the AN/A RR-66 or AN/ART-36, The length of the cable shall be 72 inches. The maximum allowable weight shall be 1.8 pounds.

3.4.6 Maintenance Fixture, Amplifier MT-2062/ARM-38 - This maintenance fixture shall be in accordance with Specification MIL-T-22163.

3.4.7 Maintenance Future, Oscillator MT-2058/ARM- 38 - This maintenance fixture shall be in accordance with Specification MIL-T-22163.

3.4.8 Maintenance Fixture, Amplifier MT-2371/ARM-46 - This maintenance fixture shall provide a means of extending the power amplifier module of RT-499/ARC-75 or ART-36 above the average height of modules in the two ratio sets. It shall include provisions for both mechanical and electrical extension. Cooling air for the power amplifier shall be provided by a 115 volt, 400 cycle, single-phase blower motor in the maintenance fixture. Overall dimensions shall not exceed 3.56 by 5 by 4.19 inches. Weight shall not exceed 2, 5 pounds.

3.4.9 Maintenance Fixture, Amplifier MT-2060/ARM-38 - This maintenance fixture shall be in accordance with Specification MIL-T-221 63.

3.4.10 Maintenance Fixture, Amplifier Generator MT-2059/ARM-38 - This maintenance fixture shall be in accordance with Spacification MIL-T-22153.

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3.4.11 Cable Assembly, Special Purpose, Electrical W-4 of AN/ARM-46 - This cable shall provide the necessary electrical connections to extend the relay unit module of Radio Receiver - Transmitter RT-499/ARC-75 to the test bench for servicing. One end of the cable shall terminate in a Cannon DB-25P or approved equivalent connector and the other end shall terminate in a Cannon DB-25S or approved equivalent connector. The length of the cable shall be 36 inches. The maximum allowable weight shall be O. 6 pounds.

3.4.12 Cable Assembly, Special Purpose, Electrical W-5 of AN/ARM-46 - This cable is provided with two double banana plugs, adapting it for use with the binding posts of Distribution Box J-11 69/ARM-46. One end of the cable shall terminate in a General Radio Company Type 274-MBSI or approved equivalent connector and the other end shall terminate in a General Radio Company Type 274- MPSSI or approved equivalent connector. The shields shall terminate in an H. H. Smith, Inc. Type 212 or approved equivalent connector. The length of the cable shall be 36 inches. The maximum allowable weight shall be O. 25 pounds.

3.4.13 Cable Assembly, Special Purpose, Electrical W-6 of AN/ARM-46 -This cable shall provide electrical connections to extend the oscillator, power amplifier, r-f amplifier, and transmitter preamplifier modules of the RT-499/ARC-75, R-823 /A RR-66, and/or T-713/ART-36 to the test bench for servicing. One end of the cable shall terminate in a Cannon DE-9P or approved equivalent connector and the other end shall terminate in a Cannon DE- 9S or approved equivalent connector. The length of the cable shall be 36 inches. The maximum allowable weight shall be O. 5 pounds.

3.4.14 Cable Assembly, Special Purpose, Electrical W-7 of AN/ARM-46 - This cable shall be used to extend the a-c power unit module of Radio-Receiver- Transmitter RT-499/ARC-75 and Radio Transmitter T-713/ART-36 to the test bench for servicing. One end of the cable shall terminate in an Amphenol 126 806 1001 or approved equivalent connector and the other end shall terminate in an Amphenol 126 807 1002 or approved equivalent connector. The length of the cable shall be 36 inches, The maximum allowable weight shall be O. 5 pounds.

3.4.15 Cable Assembly, Special Purpose, Electrical W-8 of AN/ARM-46 - This cable shall provide electrical connections to extend the 1. 85 mc i-f amplifier, 20 to 30 mc i-f amplifier, frequency shift modulator, spectrum generator and amplifier, mechanical tuning unit, limiter-discrimmator, rectifier unit, receiver r-f amplifier, transmitter preamplifier, blower and power unit modules, and relay unit module of the Radio-Receiver Transmitter RT-4 99/ARC- 75, Radio Receiver R-823 /ARR-66 or Radio Transmitter T-713/ART-36 to the test bench for servicing. One end of the cable shall terminate in a Cannon DA-15P or approved equivalent connector and the other end shall terminate in a Cannon DA- 15S or approved equivalent connector. The length of the cable shall be 36 inches. The maximum allowable weight shall be O. 5 pounds.

3.4.16 Cable Assembly, Special Purpose, Electrical (Audio) W-9 of AN/ARM-46 - This cable shall provide for the electrical connection of an a-c vacuum tube voltmeter to the AC DATA OUT jack on Distribution Box J-1169/ARM-46. One end of the cable shall terminate in a Switchcraft PJ-055 M or approved equivalent connector and the other end shall terminate in a General Radio Type 274- MB or approved equivalent connector. The length of the cable shall be 36 inches. The maximum allowable weight shall be O. 25 pounds,

3.4.17 Cable Assembly, Radio Frequency (Test-probe) W-10 of AN/ARM-46 - This cable, by means of a two and one-half inch loop of flexible conductor on one end, shall provide for the injection or extraction of signals from inductors and vacuum tubes during alignment procedures. The other end of the cable shall terminate in a UG - 886/U connector. The length of the cable shall be 52 inches. The maximum allowable weight shall be O. 2 pounds.

3.4.18 Cable Assembly, Radio Frequency (Transmitter Input W-1 1 of AN/ARM-46 - This cable shall provide electrical connection of the output of the power amplifier module to a dummy load. One end of the cable shall terminate in a UG-886/U connector and the other shall terminate in an Industrial Products 45900 or approved equivalent connector. The length of the cable shall be 60 inches. The maximum allowable weight shall be O. 2 pounds.

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3.4.19 Cable Assembly, Radio Frequency W-12 of AN/ARM-46 - This cable shall provide electrical connection between the r-f output of the power amplifier module and the chassis of radio Receiver- Transmitter RT-499/ARC-75 or Radio Transmitter T-713/ART-36 when Maintenance Fixture MT- 2371/ARM-46 is used. One end of the cable shall terminate in an Industrial Prod. with 46700 or approved equivalent connector and the other end shall terminate in an Industrial Prod. with 45900 or an approved equivalent connector, The length of the cable shall be 5 inches. The maximum allowable weight shall be O. 2 pounds.

3.4.20 Cable Assembly, Radio Frequency W-13 of AN/ARM-46 - This cable shall provide electrical connection between the output of the receiver r-f amplifier and transmitter preamplifier module in Radio Receiver-Transmitter RT-499/ARC - 75 or the output of the transmitter preamplifier module in Radio Transmitter T-713/ART-36 to the input of the power amplifier module when Maintenance Fixture MT- 2371/ARM-46 M used. Both ends of the cable shall terminate in Microdot 32-23 or approved equivalent connectors. The length of the cable shall be '7 inches. The maximum allowable weight shall be O. 1 pounds.

3.4.21 Cable Assembly, Radio Frequency W-14 of AN/ARM-46 - This cable shall provide electrical connection for r-f cables which must be lengthened when modules are extended by maintenance futures. One end of the cable shall terminate in a Microdot 32-23 or approved equivalent connector and the other end shall terminate in a Microdot 31-34 or approved equivalent connector. The length of the cable shall be 8.5 inches. The maximum allowable weight shall be O. 1 pounds.

3.4.22 Cable Assembly, Special Purpose, Electrical W-15 of AN/ARM-46 - This cable shall provide all necessary power input and control connections between Dlstribution Box J-1169(ARM-46, Receiver-Transmitter RT-499/ARC-75, and Mounting MT-2323/ARM-46. One end of the cable shall terminate in a Bendix PT06P-22-55S or approved equivalent connector, The other end shall terminate in a Bendix PT06E-10-6S or approved equivalent, and a connector capable of mating with the rear receptacle of RT-499/ARC-75. The maximum allowable weight shall be 2.3 pounds.

3.4.23 Cable Assembly, Radio Frequency, W-16 of AN/ARM-46 - This coaxial cable shall consist of a 6-foot length of RG-8A/U cable terminated on each end with a male connector UG-21D/U This cable shall provide the means of connecting the antenna terminal of the RT-499/ARC- 75, AN/ART- 36, and the AN/A RR-66 to a signal generator or a load. The weight shall not exceed 1.0 pounds.

3.4.24 Maintenance Future, Receiver, R-F Amplifier MT-2321/ARM-46 - This maintenance fixture shall provide a means of extending the R- F Amplifier module of the AN/A RR-66 above the average height of modules in the AN/A RR-66 equipment, This shall include provisions for both mechanical and electrical extension. Overall dimensions shall not exceed 5 by 4- 1/2 by 1-1/2 inches. Weight shall not exceed 1,4 pounds.

3,4.25 Maintenance Fixture, Transmitter Preamplifier MT-2322/ARM-46 - This maintenance fixture shall provide a means of extending the transmitter-preamplifier module of the AN/ART- 36 above the average height of modules in the AN/ART-36 equipment. This shall include provisions for both mechanical and electrical extension. Overall dimensions shall not exceed 5 by 4-1/2 by 1-1/2 inches. Weight shall not exceed 1.4 pounds.

3.4.26 Tool Kit TK-(*)/V - Each test set shall include a kit of tools which shall consist of the following:

(1) 1 - No, 8 Bristo Wrench

(2) 1- "T" Handle Wrench for No. 8 Hex Socket Cap Screw

(3) 1 - Tuning Tool with screwdriver on one end and slotted fitting on opposite end.

(4) 1 - Alignment tool with plastic handle, insulated steel shaft and steel tip

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(5) 1- No. 4 Bristo Wrench

(6) 1- Right angle Phillips screwdriver with No, 1 point at each end.

(7) 1- No. 6 Bristo Wrench

(8) 1- Flexible shaft trimmer adjusting tool

(9) 1- Adjusting Tool

(lo) 1- Tuning Wrench

(11) 1- Feedthrough Connector Wrench

(12) 1- Front Panel Connector Wrench

(13) 1- Tuning Wand

(14) 1- Tuning Tool

(15) 1- Extractor Tool - Tube

(16) 1- Actuator - Switch

(17) 1 Tool - Adapter

3.4.27 Adapter Kit MK- (*)/U - Each test set shall include an adapter kut which shall consist of the following:

(1) 1 - Adapter Connector to permit coupling of a male miniature r-f connector, of a type commonly used m the AN/ARC-75, AN/ART-36 and AN/A RR-66 to a female BNC type UG-290A/U connector.

(2) 1 - UG-201A/U Adapter

(3) 1 - Adapter connector to permit coupling of a male miniature r-f connector, of a type commonly used in the AN/ARC-75, AN/ART-36 and AN/A R.R-66 to a male BNC type UG-88A/U Connector.

3.4.28 Tuning Cover Kit MK-(*)/ARM-46 - Each test set shall include a tuning cover kit which shall consist of the following:

(1) 1- Power Amplifier Module Tuning Cover.

(2) 1- Preamplifier Module Tuning Cover

(3) 1- Receiver Module Tuning Cover

(4) 1- Spectrum Generator Module Tuning Cover

(5) 1- Oscillator Module Tuning Cover

(6) 1- R-F Amplifier Preamplifier Module Tuning Cover

(7) 1- Transmitter Preamplifier Module Tuning Cover

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval. This data shall 'elude ª list 'f all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. This data may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-T-21200. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract, Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422, modifying the test limits of the applicable tests to correspond with the limits specified herein. Interference tests shall be conducted in accordance with Specification MIL-I-6181 Drop tests for transit and combination cases shall be in accordance with Speci - fication MIL-T-21200.

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4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality be equivalent to the approved preproduction sample. Each equipment shall be capable of successful y passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection or if already accepted by the government It shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests -When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the approved preproduction sample in design, construction, workmanship, performance and quality and that it is capable of meeting all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector, Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

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4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability y Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws -an-d imperfect workman - ship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10- minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment, A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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4.4.2 Sampling Tests - sampling tests shall be conducted on equipments which have successfully passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	I
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be Subjected to the following tests

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environment.al conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (See para. 4. 3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

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4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity, Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests, As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Speci - fication MIL- T- 18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the con- tractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

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5. PREPARATION FOR DELIVERY

5.1 General -All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555,

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 115 ± 1.0 V AC and 27.5 ± 0.5 V DC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.2 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply.

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall hare precedence over all applicable subsidiary speci - fications referenced therein. All referenced specifications shall apply to the extent specified.

6. 3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-T-21200, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

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6.4 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby recurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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